統計實習總結

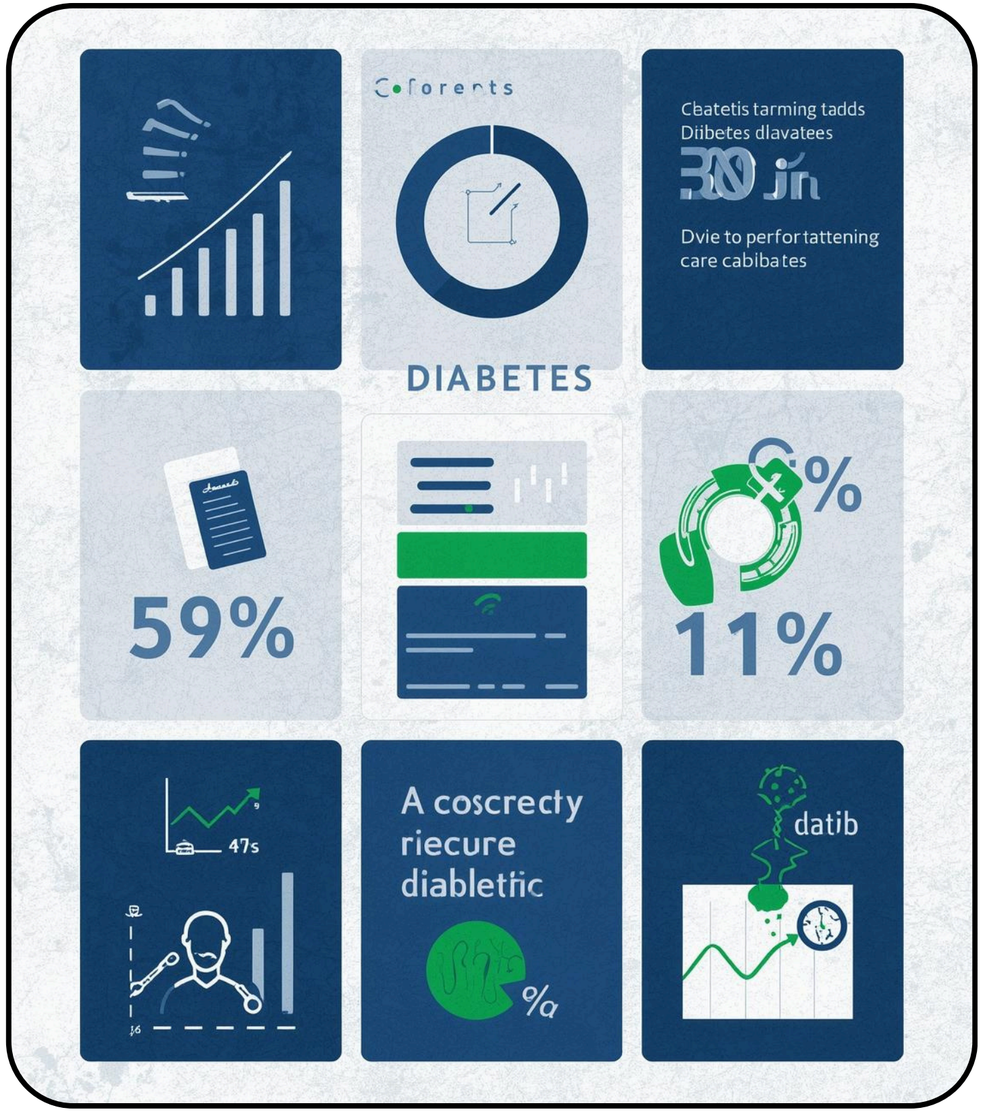
第二組

工管二乙 113370450 蘇孟孜

工管二乙 113370228 吳彥寬

工管二乙 113370220 游子涵

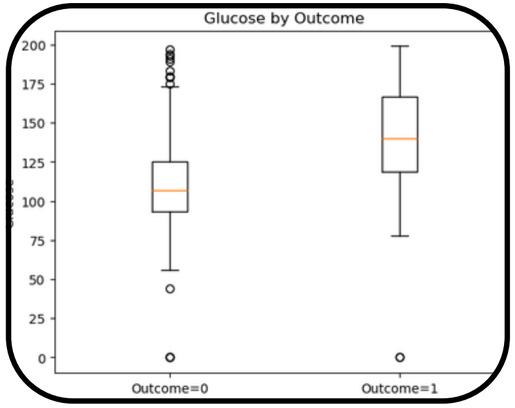
工管三甲 112370128 陳冠維

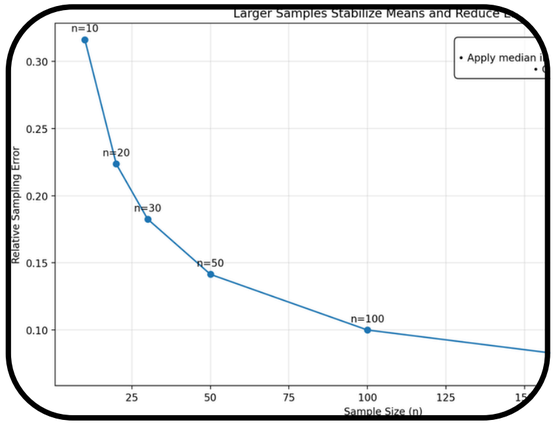


Core Project Overview

```
Glucose                 5
BloodPressure          35
SkinThickness         227
Insulin               374
BMI                    11
dtype: int64
```

nThickness	Insulin	BMI
35	0	33.6
29	0	26.6
0	0	23.3
23	94	28.1
35	168	43.1
0	0	25.6
32	88	31
0	0	35.3
45	543	30.5
0	0	





1. Project Overview & Objectives
   - Analyze Diabetes.csv with 768 entries and 9 key physiological variables.
   - Identify diabetes risk factors and evaluate how sample size impacts accuracy.

2. Data Quality: The "Hidden" Missing Values
   - Initial Check: The isna().sum() function returned 0, suggesting no missing values.
   - The Discovery: Several physiological variables (e.g., Glucose, BMI, BloodPressure) contained "0" values, which are biologically impossible.

3. Key Visualization Insights
   - Boxplots and Histograms confirm significantly higher glucose levels in patients.
   - Scatter plots link Age and BMI, highlighting specific high-risk clusters.

4. Statistical Findings & Improvements
   - Confirmed that larger samples n effectively stabilize means and reduce errors.
   - Future work includes median imputation for "0" values andworkflow optimization.

實習核心總覽

```
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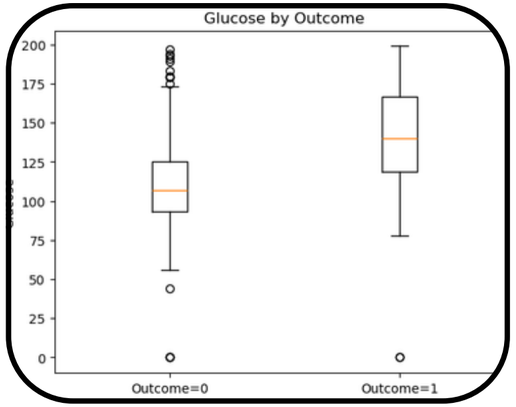

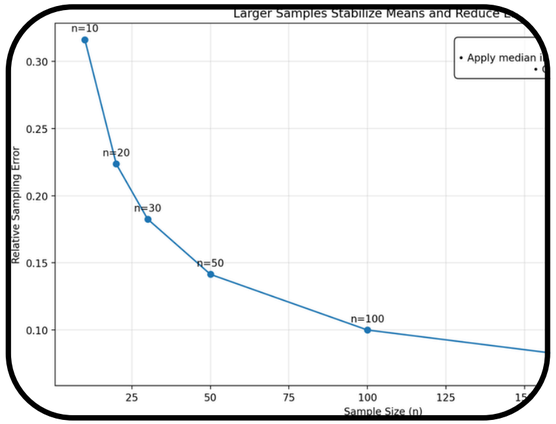

1. 資料概況

本次實習以 Diabetes.csv 為分析資料來源，共包含 768 筆樣本 與多個健康相關變數，可用來探討糖尿病與各項指標之間的關係。

2. 資料品質檢查

資料表面上沒有缺漏值，但部分欄位出現不合理的 0 值，代表可能存在隱藏缺失，分析時需特別注意。

3. 主要分析發現

從圖表與分組比較可發現，Glucose 與 BMI 呈現右偏分布，且糖尿病組的血糖值整體明顯較高，顯示 Glucose 與 Outcome 關聯最明顯。

4. 抽樣模擬結果

透過 Bootstrap 抽樣模擬 可觀察到，樣本平均數大多集中在母體平均附近，顯示平均數具有一定穩定性，也幫助理解抽樣分配的基本概念。

結論與學習成果分析

本次實習強調Glucose的重要性，並提升統計推論及資料分析的實務能力。